Why a fair-testing paper is our best next step

GBLSTSVM fuzzy study · Jayagna · 4 November 2026 · a short summary in plain words

The short answer

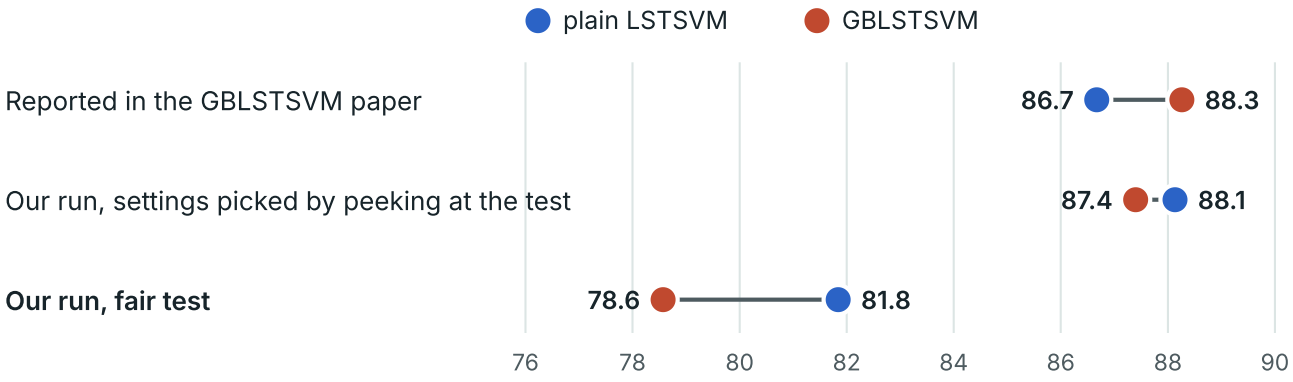
- **We looked hard for a new fuzzy weighting rule for GBLSTSVM.** Every rule we found or invented was either already published, or no better than a simple idea from 1985.
- **Along the way we found something nobody has reported.** When plain LSTSVM and GBLSTSVM are tested fairly, plain LSTSVM wins. GBLSTSVM only looks better when its settings are chosen by looking at the test answers.
- **That finding is new, our experiments already back it up, and it still lets us present a new fuzzy model.** It is a safer and stronger paper than a new rule that does not beat the old ones.

1. What "fair testing" means

Think of two students getting ready for an exam. Each can choose how to study. A **fair** teacher lets them practise only on practice papers, then gives the real exam once. An **unfair** teacher lets them try every way of studying, marks each try against the real exam, and keeps the best mark. The second student's mark looks great, but it says little about how they would do on a new exam.

A model has settings too, such as how strict it is and how the balls are built. Choosing those settings by looking at the test score is the unfair teacher. The more settings a model has, the more it gains: GBLSTSVM has 490 combinations to try, plain LSTSVM has 98.

Average accuracy (%) on data with no flipped labels



Our runs: 21 datasets, each split 10 different ways. The paper used 34 datasets, so its row is a strong hint, not proof. With 30% of labels flipped, the fair test gives 67.0 vs 64.0, while peeking gives 84.0 vs 84.2.

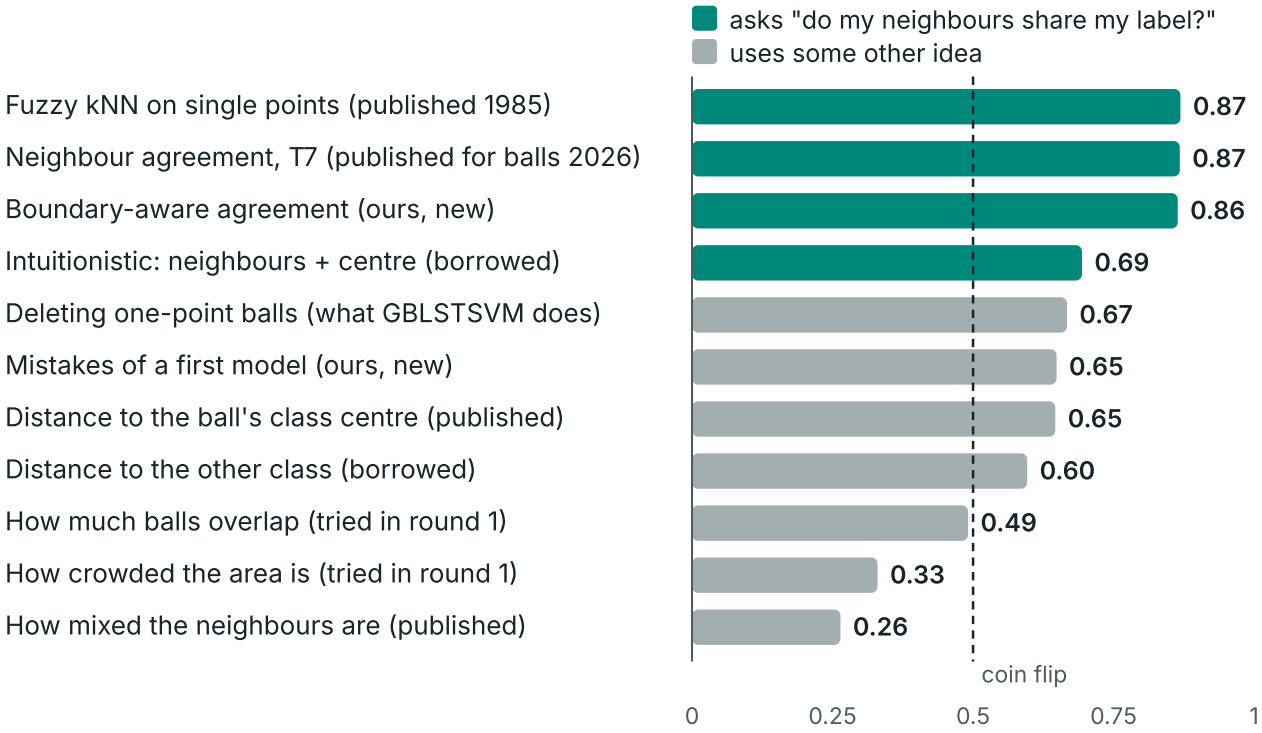
Peeking lifts both models by 6 to 20 points and closes the gap between them, and the paper's numbers sit right next to our peeking numbers. Tested fairly, plain LSTSVM wins at every noise level, by 3 to 4 points.

2. Why not just invent a new fuzzy rule?

GBLSTSVM groups nearby points into **balls**. A fuzzy weighting rule gives each ball a trust score between 0 and 1, so that balls with wrong labels count less. So a good rule must be able to tell which balls carry wrong labels.

We measured exactly that. We flipped 20% of the labels on purpose and checked how well each rule gives low trust to the flipped ones. A score of 1 is perfect; 0.5 is no better than tossing a coin.

How well each rule spots wrong labels (20% of labels flipped)



Average over 21 datasets. "Ours, new" are the two rules we invented this week; "borrowed" are rules from other fuzzy SVM papers that nobody had used on GBLSTSVM.

1 **Only one idea works, and it is taken.** Rules built only on "do my neighbours share my label?" score 0.86–0.87. Every other idea scores 0.67 or less, some do worse than a coin, and mixing one in lowers the score. The neighbour idea was published in 1985 and used on balls in 2026.

2 **GBLSTSVM leaves little to fix.** Before any weights are used, it throws away every ball that holds a single point. That already removes 82–88% of the wrong labels, so over 99% of the balls left are clean. In our first round, every weighting rule moved accuracy by about 1 point or less.

3 **A careful reviewer would test it fairly.** Our best new rule only ties the 1985 idea. A reviewer who tests it the fair way, as we did, would find no real gain over the old idea.

This week alone we tried four more rules, two of our own and two borrowed. None beat the old neighbour idea.

3. Why the fair-testing paper is the best move now

This is the paper we can write. Each of these four findings is new as far as we can find:

1. **A hidden clean-up step.** Throwing away one-point balls quietly removes 82–88% of the wrong labels, but also 41–60% of the correct points, mostly points near the border between the classes and points from the smaller class.
2. **Fair testing flips the ranking.** Plain LSTSVM beats GBLSTSVM when both are tested fairly. The published ranking comes back only when settings are picked using the test answers.
3. **One fair comparison of fuzzy rules.** About ten weighting rules from old and new papers, all put on GBLSTSVM and all tested the same way. It tells other researchers which ideas work and which do not.
4. **A new fuzzy GBLSTSVM.** Keep the one-point balls and give each ball a trust score from its neighbours (we call it keep + T7). Tested fairly, it beats GBLSTSVM by 2.6–4.9 points and the fuzzy model GBFTSVM by 3.7–5.6 points. The trust rule is known; using it this way in GBLSTSVM is new, the same kind of novelty that GBFTSVM itself claims.

Option A: keep hunting for a new fuzzy rule

- Needs a rule that beats a 1985 idea. We tried about ten; none did.
- More months of work with no sign of a gain.
- A fair reviewer is likely to find the gain disappears.

Option B: write the fair-testing paper

- The experiments are done: 21 datasets, 10 splits each, 4 noise levels, careful statistics.
- Four new findings, including a new fuzzy model.
- Gives the lab a fair test for its next ball-based models, before reviewers apply one.

Papers like this are respected. Demšar's 2006 paper on how to compare classifiers fairly is one of the most cited papers on machine learning methods, and "Do we need hundreds of classifiers?" (2014) is widely cited for the same kind of careful re-testing.

Working title: "Where do granular balls help least squares twin SVMs? A fair evaluation and a simple fix." All code and results are on GitHub (Jayagna/GBLSTSVM).